A distributed embedded system for real-time monitoring, analysis, anomaly detection, and intelligent decision support

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Smart Indoor Environmental Intelligence System (SIEIS)

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Problem Statement

Silent Deterioration: Indoor environments (classrooms, labs) degrade without immediate human awareness (e.g., rising CO_2, TVOCs).

Reactive, Not Proactive: Occupants only react when discomfort is extreme ("stuffy room"), often after cognitive or health impacts occur.

Dumb Monitoring: Current cheap systems only display raw numbers (e.g., "CO_2 = 1100 ppm") without context or actionable advice.

Missing Intelligence: Lack of automated trend analysis to predict deterioration before it becomes a problem.

Proposed Solution

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THE SOLUTION (SIEIS):

- An edge-based sensor node utilizing an ESP32 to acquire multi-parameter environmental data.
- A robust I²C/ADC hardware architecture to digitize conditions.
- A centralized software backend (Python/SQLite) for data storage and processing.
- An intelligence layer providing scoring, anomaly detection, and decision support via a web dashboard.

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Objectives

KEY OBJECTIVES:

- **Monitor**: Capture Temp, Humidity, Pressure, eCO_2, TVOC, AQI, Light, and Noise accurately.
- **Transmit**: Send validated JSON data packets reliably over Wi-Fi (HTTP).
- **Analyze**: Calculate Environmental Health and Comfort scores locally.
- **Decide**: Implement rule-based intelligence for early warning recommendations.

System Architecture

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BME280
(I²C)

ENS160
(I²C)

BH1750
(I²C)

BME280
(I²C)

ESP32 Microcontroller
Acquisition | Validation | JSON Fornatting

Backend Server
Python / Flask / SQLite

Intelligence Layer
Scoring | Trend Analysis | Prediction

Web Dashboard (Decision Support & Alerts)

06 Hardware & Parameters Measured

Component	Interface	Parameters Measured / Role
ESP32	GPIO / Wi-Fi	Master Controller, Sampling, Wi-Fi Gateway
BME280	I²C	Temperature, Humidity, Atmospheric Pressure
ENS160	I²C	TVOC, eCO_2 Estimate, Air Quality Index (AQI)
BH1750	I²C	Light Intensity (Lux)
Sound Sensor	Analog (ADC)	Sound Intensity / Ambient Noise Level
OLED (Optional)	I²C	Standalone Local Diagnostic Display
Power / Chassis	USB 5V	Hardware Power, Prototyping Board, Enclosure

Technologies / Tech Stack

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Hardware & Edge

- **Platform:** ESP32 Development Board
- **Sensors:** BME280, ENS160, BH1750, Analog Mic
- **Protocols:** I²C, ADC, Wi-Fi
- **Language:** C++ (Arduino/PlatformIO)

Backend & Data

- **Language:** Python
- **Framework:** Flask / FastAPI (HTTP API)
- **Database:** SQLite (Time-series storage)
- **Intelligence:** Python (Rule-based & Statistical logic)

Dashboard UI

- **Structure:** HTML5, CSS3
- **Logic:** JavaScript (Vanilla)
- **Visualization:** Chart.js (Real-time graphs)

08 Budget & Resource Allocation

Component	Target Budget (₹)	Allocation Purpose
ESP32 Controller	₹300 – ₹400	Data acquisition, I²C bus master, Wi-Fi POST requests
BME280 Sensor	₹250 – ₹350	Precision thermal comfort & pressure measurement
ENS160 Sensor	₹700 – ₹900	TVOC, eCO_2 estimate, and gas-air quality indication
BH1750 Light Sensor	₹120 – ₹180	Lux level monitoring for lighting adequacy
Sound Sensor Module	₹80 – ₹120	Ambient activity & noise level classification
Breadboard & Jumpers	₹200 – ₹300	Non-destructive prototyping & interconnects
USB Cable & Power	₹100 – ₹200	Reliable 5V power supply & data flashing
Misc (Enclosure / OLED / Spares)	₹200 – ₹300	Casing, diagnostic OLED, and component backup
Total Estimated Cost	**₹1,950 – ₹2,750**	**Fully compliant with ₹3,000 project budget**

Phases:

Phase	Task
Hardware	Component procurement & basic breadboard assembly
Firmware	I²C interfacing (BME280) & reading basic data
Firmware	Integrate ENS160, BH1750, and sound sensor
Communication	Establish ESP32 Wi-Fi connection & JSON formatting
Backend	Python server to receive data & SQLite integration
Intelligence	Implement Health Score & Trend Analysis algorithms
Dashboard	Develop UI with live Chart.js visualization

Expected Results & Future Scope

Expected Output (Final Review)

- A fully functional, stable hardware node transmitting data continuously.
- A live dashboard displaying real-time metrics and historical trends.
- Demonstrable intelligence: the system reacting to controlled environmental changes (e.g., simulating poor ventilation to trigger a trend alert).
- A resilient embedded architecture with proper error handling.

Future Extensions (Post-V1)

- **Multi-Node Network:** Deploying multiple ESP32s (e.g., Classroom, Lab) to a single centralized dashboard.
- **Adaptive Baselines:** System learning 'normal' room conditions over time to refine anomaly detection.
- **MQTT Integration:** Upgrading from HTTP to MQTT for lighter, publish-subscribe networking.
- **Local OLED:** Adding a small I²C display to the sensor node for offline status verification.

11 Novelty & Uniqueness of SIEIS

Feature / Dimension	Traditional IoT Environmental Monitor	SIEIS (Proposed System)
Primary Output	Raw scalar sensor values	Contextual Health Score, Comfort Index & Trends
Data Integrity	Accepts raw sensor noise / bad packets	Real-time range & rate-of-change validation
Alert Triggering	Static single-parameter threshold breach	Multi-parameter synthesis & predictive forecasting
User Guidance	Passive dashboard display	Direct, actionable recommendations
Architecture	Simple sensor polling loop	Modular Embedded State Machine + Python Intelligence Engine



Thank you!

LET US KNOW IF YOU HAVE QUESTIONS OR CLARIFICATIONS.